

Financial News for Exchange Rate Predication

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Abstract—Currency is a critical medium of exchange that connects individuals to the global economy. However, the exchange rate of national currencies is dynamic and unpredictable. Previous research has demonstrated that financial news plays a key role in forecasting market trends. Therefore, a considerable amount of financial data is used in exchange rate prediction studies. However, these data sources are often limited or outdated, and the potential of using multisource financial news data remains underexplored. This study utilizes news from multiple United States financial news networks to identify the optimal combination of multisource data by integrating various text representations, sentiment features, and machine learning (ML) techniques for exchange rate prediction. The findings reveal that combining support vector machine (SVM) and term frequency-inverse document frequency (TF-IDF) yields the best accurate in ML-based exchange rate forecasting using multisource financial news.

Keywords—exchange rate predication, financial news, text mining, machine learning

I. INTRODUCTION

The foreign exchange market is the world's largest and most liquid financial market, with an average daily trading volume of \$5 trillion [1]. This market is not governed by a single institution or country, which renders exchange rates highly volatile. Early studies by Meese and Rogoff [2] argued pessimistically that long-term exchange rate trends could only be predicted using a simple random walk, suggesting that overcoming this limitation in the forex market would be challenging. However, with the rise of artificial intelligence and the increasing availability of numerical and textual data online, researchers are increasingly turning to big data approaches to improve exchange rate predictions and address these challenges.

Over the years, various analytical concepts and methods have been applied to the foreign exchange market. Traditional economic approaches, including fundamental analysis, technical analysis, and econometric models, have been used to predict exchange rates [3-5]. However, these methods often struggle with the nonlinear nature of exchange rate predictions, especially under high-frequency conditions [6]. With advancements in machine learning (ML) technologies and markedly breakthroughs across various fields, economists have

begun integrating ML techniques with traditional econometric models to improve predictive performance.

Historically, data and fundamental information have been standard reference points in economics. However, recent studies have highlighted the importance of financial news as an alternative information source for simulating market trends [7]. The information and sentiment conveyed in news reports considerably influence financial markets [8], particularly for market participants and brokers who process large volumes of information. Real-time financial news, whether positive or negative, can directly influence investors' and multinational enterprises' decision-making. In addition, studies have shown that different news outlets may exhibit biases in their reporting [9], potentially influencing market participants' decisions in different ways.

II. LITERATURE REVIEW

Gillam et al. [10] emphasized that the hidden information in news reports is a key element in shaping investment strategies, regardless of the source of the information. Xing et al. [7] also highlighted the role of financial news in simulating market behavior and making predictions, although most research has been focused on the stock market.

Reviewing previous studies on learning methods and text feature extraction, Peramunetilleke and Wong [11] used news headlines to predict intraday exchange rate trends. By performing statistical analysis and generating rule-based algorithms on the keywords in each headline, they found that term frequency (TF) combined with category discrimination yielded better results in exchange rate prediction. Patil et al. [12] utilized Reuters news and applied the TF-inverse document frequency (TF-IDF) text representation method, testing several ML models, including support vector machine (SVM), K-nearest neighbors (KNN), and Naïve Bayes (NB), with SVM achieving the highest prediction accuracy. Sato et al. [13] used news from SMBC Trust Bank and Reuters, extracting features via TF-IDF and integrating them with an SVM classifier to predict exchange rates for an automated trading mechanism.

Semiromi et al. [14] pointed out that earlier exchange rate prediction studies using financial texts overlooked the influence of events in economic calendars. By incorporating economic calendar data and developing a new sentiment lexicon tailored for the foreign exchange market, they tested their method with various classifiers, finding that extreme gradient boosting (XGBoost) produced the best results. Lee et al. [15] demonstrated that incorporating news sentiment analysis reduced the model's root mean squared error and mean absolute percentage error. Chen et al. [16] compiled a large financial news corpus, categorizing it by time, topic, and type. They used a Bert-based hierarchical aggregation model (BHAM) to predict exchange rate trends, showing that BHAM combined with news categorized by type outperformed SVM in predictive accuracy. Xing et al. [7] applied FinBERT to extract sentiment features from news and used a multilayer perceptron model for foreign exchange prediction, surpassing the performance of NB. Yono et al. [17] found that trending topics and their associated sentiments were crucial for shaping investment strategies, with SVM outperforming bidirectional long short-term memory.

However, the combination of different text representation methods and sentiment features with supervised learning methods yields varying results in classification models, making it necessary to conduct experiments to assess their strengths and weaknesses. Additionally, most related studies focus on single-source news data or datasets from open platforms, without exploring multisource news to identify the best combinations or investigating whether different news sources yield similar prediction accuracies for a single currency.

This study seeks to address the use of multisource news by applying various text representation methods, sentiment features, and ML classifiers to identify the best combination for predicting exchange rate trends. It also aims to understand the impact of different news sources on exchange rate predictions.

III. EXPERIMENT

The experiment in this study is divided into six stages: (1) data collection, (2) data preprocessing, (3) feature extraction, (4) training and testing split, (5) model development, and (6) evaluation, as shown in Fig. 1.

Financial news articles were crawled from multiple financial news websites, while currency exchange rate data was collected. The financial texts underwent preprocessing, and the exchange rate data was labeled (rise/fall) on a daily basis. The financial texts were then represented as vectors using various text feature extraction methods, including TF-IDF, Word2vec, and bidirectional encoder representations from Transformers (BERT). ML classifiers such as SVM, KNN, NB, and XGBoost were selected, and sentiment features were extracted from the texts using the Loughran-McDonald financial sentiment dictionaries (LMFD)¹. The model evaluation metrics used were accuracy, F1 score, and area under curve (AUC).

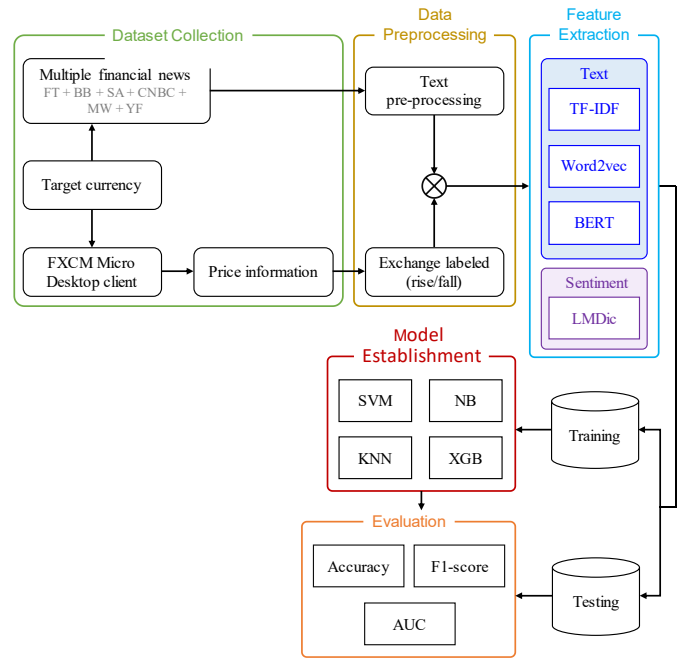


Fig. 1. Experimental setup and architecture

A. Dataset collection

This study, based on a survey by the Society for Worldwide Interbank Financial Telecommunication (SWIFT)², identifies the top 20 payment currencies and their respective markets. The most widely circulated currency is the U.S. dollar, so the study uses the U.S. dollar as the base currency. It then selects one widely circulated currency from Europe, Asia, and Australia as the prediction targets. Ultimately, the study focuses on EUR/USD, JPY/USD, and AUD/USD as the target pairs for trend forecasting.

The research uses the News API³ from Event Registry to gather financial news related to these currencies over a five-year period, from January 1, 2016, to December 31, 2020. The sources include six prominent financial news outlets: Financial Times, Bloomberg Business, Seeking Alpha (SA), CNBC, MarketWatch (MW), and Yahoo Finance. According to a survey by eBusiness, these outlets were ranked among the top 15 most popular business and financial news platforms⁴. Additionally, to ensure direct relevance to the target currencies, only articles that mention the target currencies in their titles or content are included in the analysis.

B. Data preprocessing

The news data collected through the News API is organized into fields such as news titles, content, sources, and dates during the data collection process. However, the raw news text contains many unusable elements, such as webpage links, punctuation, and special characters. To convert this unstructured financial text into clean, preprocessed data, a text preprocessing procedure is necessary. Given the nature of TF-IDF, it is also

¹ <https://github.com/hanzhichao2000/pysentimentv>

² Society for Worldwide Interbank Financial Telecommunication (SWIFT): <https://www.swift.com>

³ News API: <https://www.newsapi.ai/>

⁴ top 15 most popular business and financial news platforms: <http://www.ebizmba.com/articles/business-websites>

important to remove less meaningful words, such as articles and prepositions, to reduce their impact on the TF-IDF results.

For the exchange rate data, daily closing prices, opening prices, lowest prices, and highest prices of the target currency pairs are collected using the FXCM Micro Desktop client provided by FXCM⁵. Because the study focuses on forecasting foreign exchange trends, factors such as different brokers' spreads and overnight swap rates are temporarily disregarded for simplification.

The study uses financial news published on the previous day to predict the exchange rate trends for the following day. To facilitate model training, each news article is assigned a label (i.e., the next day's exchange rate trend). This study employs the total daily return (close-to-close) method to calculate exchange rate returns. In this method, the next day's closing price is subtracted from the current day's closing price. If the result is negative, it is classified as a decline (category 0); otherwise, it is classified as an increase (category 1). This method is used to generate the answer labels. Additionally, to account for the impact of holiday news on market reopening, the labels for news published during market closures are determined by subtracting the last closing price of the holiday period from the first closing price after the market reopens.

C. Text feature extraction

This study utilizes three text representation methods to extract features: (1) TF-IDF, (2) Word2Vec, and (3) BERT.

Given the large size of the experimental dataset, with a unique vocabulary exceeding 250,000 words, memory limitations and extended training times are potential challenges. Referring to Yu [18], it was found that reducing the dimensions of TF-IDF features to 10% based on word frequency maintained strong feature performance while effectively minimizing training times. Additionally, because TF-IDF generates features for the entire text, the resulting feature vectors do not require further processing after calculation.

In contrast, Word2Vec and BERT create word-level vectors. To address this, the study follows Kilimci et al. [19], who demonstrated that averaging word embedding vectors to form document vectors is an effective approach. Therefore, the Word2Vec and BERT word vectors are averaged to represent the vectorized form of the entire news article, enabling their application in ML classifiers.

D. Sentiment feature extraction

Many studies [7, 20-22] suggest that incorporating sentiment features can improve prediction performance, making the inclusion of sentiment features an essential aspect of this study. Rather than focusing on the emotions conveyed by individual sentences or words, this study emphasizes the overall positive or negative sentiment of the article as a whole.

To extract sentiment features from the articles, the study uses the LMFD. The number of positive and negative words in each article is calculated to determine its sentiment polarity. The method for calculating sentiment is outlined in Equation 1, where Pos represents the number of positive words and Neg

represents the number of negative words. This approach ensures a comprehensive analysis of the sentiment expressed in financial news articles.

$$\text{Sentiment_polarity} = \frac{\text{Pos} - \text{Neg}}{\text{Pos} + \text{Neg}} \quad (1)$$

E. Training and testing data

Due to the time-series nature of the news data, employing k-fold cross-validation would disrupt the sequential relationship within the data. Therefore, this study adopts the sliding window method, as illustrated in Figure 2. The preprocessed dataset is divided into three subsets based on the sliding window concept, with each subset containing both training and testing sets. The final prediction results are obtained by averaging the outcomes from the testing sets.

For example, in the first sliding window subset, the training data spans three years (January 1, 2016-December 31, 2018), while the subsequent year (January 1, 2019-December 31, 2019) serves as the testing data. In the next subset, the sliding window shifts the entire time range backward by six months. That is, the training data for the second subset covers July 1, 2016-June 30, 2019, with the testing data spanning July 1, 2019-June 30, 2020. This ensures that the duration of all datasets remains consistent, and the process continues similarly for subsequent subsets.

IV. EXPERIMENTAL RESULT

In the experiments, various combinations of ML models and text representation modules are evaluated, as follows: **SW** (SVM + Word2vec), **SB** (SVM + BERT), **ST** (SVM + TF-IDF), **KW** (KNN + Word2vec), **KB** (KNN + BERT), **KT** (KNN + TF-IDF), **NW** (NB + Word2vec), **NB** (NB + BERT), **NT** (NB + TF-IDF), **XW** (XGBoost + Word2vec), **XB** (XGBoost + BERT), and **XT** (XGBoost + TF-IDF).

As illustrated in Fig. 2, the experimental results show that sentiment features have a more significant positive impact on most combination methods when predicting the Australian dollar, compared to other currencies. For the Japanese yen, sentiment features also lead to improvements in most methods, though the effect is less pronounced than for the Australian dollar. In contrast, the addition of sentiment features has minimal impact on the performance of combination methods for the euro, with only slight increases or decreases observed in a few methods. Therefore, the influence of sentiment features on different currencies is not consistent. To clarify their effects across various classifiers, subsequent discussions will focus on sentiment features integrated with text representation methods, primarily analyzed through classifiers.

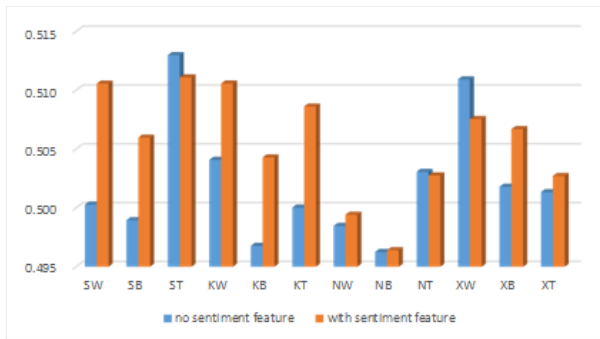
SVM shows better performance when paired with TF-IDF as the text representation method, compared to Word2Vec and BERT. The AUC values before and after the addition of sentiment features show slight improvements in most combinations. Initially, SVM performed less effectively with the Word2Vec and BERT combinations, but after incorporating sentiment features, these combinations exhibited significant improvements in predictive performance. Although Word2Vec

⁵ FXCM: <https://www.fuhuimkt.com/tc/>

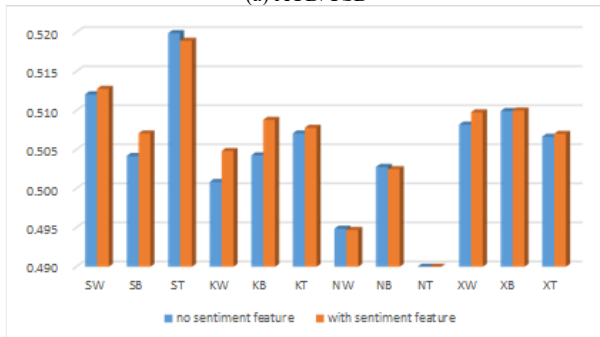
and BERT combinations with sentiment features still underperform compared to TF-IDF-based combinations, it is clear that sentiment indicators positively contribute to SVM's ability to learn dense vector representations. This underscores the beneficial impact of sentiment features on SVM when working with dense vectors.

KNN-based combinations, whether incorporating sentiment features or not, consistently underperform compared to SVM-based combinations. However, sentiment features have a relatively significant impact on KNN. Nearly all combinations involving KNN show positive performance improvements when sentiment features are added. Therefore, sentiment features are generally beneficial in enhancing the classification performance of KNN-based combinations.

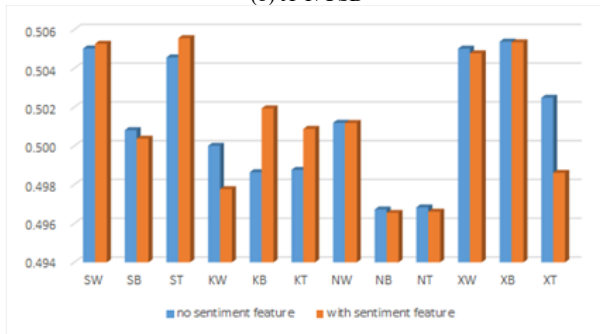
The NB classifier demonstrates the poorest performance among all classifiers in this study. Despite its computational efficiency, most combinations based on NB exhibit predictive performance worse than random walks. Moreover, the addition of sentiment features does not result in significant changes to the AUC values across the combinations.



(a) AUD/USD



(b) JPY/USD



(c) EUR/USD

Fig. 2. Experimental results and analysis

However, the effect of adding sentiment features on XGBoost remains unclear. For many combinations, there is no notable change or evident trend in AUC after incorporating sentiment features. This suggests that sentiment features do not consistently influence the performance of XGBoost-based models in a significant or predictable manner.

V. CONCLUSION

Based on the AUC analysis in the experiments, SVM + TF-IDF emerged as the best-performing combination for predicting exchange rate trends across different currencies. Even with the integration of sentiment features, SVM + TF-IDF consistently remained the second-best choice and proved to be the optimal method for predicting trends in the euro. This finding aligns with past research, where the combination of SVM and TF-IDF has been commonly favored for its effectiveness.

Although previous studies suggest that ensemble learning methods can outperform single classifiers, XGBoost did not meet expectations within the context of this research framework. This discrepancy may be due to the use of fixed parameters across all data sources and currency types (with default values in the experiments). XGBoost has over 10 tunable parameters, and research in other fields shows that it tends to deliver better results after parameter tuning, such as with methods such as GridSearch. Therefore, under the conditions of analyzing currency-specific news without parameter adjustments, SVM demonstrated relatively superior performance.

Additionally, this study focused solely on ML methods and did not explore the potential of deep learning approaches. Thus, comparing deep learning methods represents a promising avenue for further research.

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